RANDOMIZED SMOOTHING AT HIGH CONFIDENCE: THE GAUSSIAN IS OPTIMAL FOR EVERY NOISE LAW

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Abstract. Randomized smoothing certifies a classifier by averaging a base classifier over a noise law q on $\mathbb{R}^d$. We consider certificates that hold for every base classifier and require only that the top class has probability at least $1-\alpha$ under the noise. We prove that if $\mathbb{E}\|\eta\|_2^2 \le d\sigma^2$ for $\eta \sim q$, then for $0 < \alpha < 1/4$ and $d \ge 1012\,\alpha^{-2}\log(16/\alpha)$ no such certificate exceeds $8\sigma\sqrt{\log(16/\alpha)}\,d^{1/p-1/2}$ in the ℓ_p norm, $2 \le p \le \infty$. The Gaussian $N(0,\sigma^2 I)$ certifies $\sigma\Phi^{-1}(1-\alpha)\,d^{1/p-1/2}$, so it is optimal up to an absolute constant factor at every confidence level. Earlier bounds for general noise laws allow the certified radius at fixed noise level to grow like $\alpha^{-1/2}$ as $\alpha \to 0$; the correct rate is $(\log 1/\alpha)^{1/2}$. The proof uses only the second moment of q: no product structure, symmetry, density or log-concavity.

1. Introduction

Let $f\colon \mathbb{R}^d \to \mathcal{Y}$ be a measurable classifier and let q be a Borel probability measure on $\mathbb{R}^d$. The smoothed classifier $g(x) = \arg\max_c \mathbb{P}_{\eta\sim q}(f(x+\eta)=c)$ is the object of randomized smoothing [4, 5, 2]. For the Gaussian $q = N(0,\sigma^2 I)$, Cohen, Rosenfeld and Kolter [2] showed by the Neyman–Pearson lemma that if the top class A has probability $p_A > 1/2$ at x, then $g(x+\delta) = A$ whenever $\|\delta\|_2 < \sigma\Phi^{-1}(p_A)$, and that this radius cannot be improved using p_A alone. Since the ℓ_∞ ball of radius r lies in the ℓ_2 ball of radius $r\sqrt{d}$, the same noise certifies the ℓ_∞ radius $\sigma\Phi^{-1}(p_A)/\sqrt{d}$.

Whether another noise law does better for ℓ_p, $p > 2$, has been studied from several directions. Blum, Dick, Manoj and Zhang [1] proved that a noise law which certifies ℓ_p radius ε for every base classifier with score gap δ must have $\mathbb{E}\eta_i^2 = \Omega(d^{1-2/p}\varepsilon^2(1-\delta)/\delta^2)$ for most coordinates i. Kumar, Levine, Goldstein and Feizi [3] showed that for a large class of product noise laws the certified ℓ_p radius decreases as $O(d^{1/p-1/2})$. Yang et al. [8] obtained a bound of the same order in d for a broad class of smoothing schemes that use only the probability of the top class, including schemes whose noise depends on the input, and Wu, Bojchevski, Kuvshinov and Günnemann [7] gave upper bounds of order $d^{-1/2}$ for the ℓ_2 radius under continuous symmetric noise.

Read at a fixed noise level σ per coordinate and confidence $p_A = 1-\alpha$, so that the score gap is $1-2\alpha$, the bound of [1] gives $\varepsilon = O(\sigma d^{1/p-1/2}\alpha^{-1/2})$, while the Gaussian certifies $\sigma\Phi^{-1}(1-\alpha)d^{1/p-1/2} \sim \sigma\sqrt{2\log(1/\alpha)}\,d^{1/p-1/2}$. The gap between $\alpha^{-1/2}$ and $(\log 1/\alpha)^{1/2}$ is the question of whether a non-Gaussian law can buy more radius at high confidence. The theorem below shows it cannot.

Definition 1.1. Let $\|\cdot\|$ be a norm on $\mathbb{R}^d$ and $0 < \alpha < 1/2$. The *certified radius* $R_q(\alpha, \|\cdot\|)$ is the supremum of $\varepsilon \ge 0$ with the following property: for every measurable $f\colon \mathbb{R}^d \to \mathcal{Y}$, every $x \in \mathbb{R}^d$ and every class A with $\mathbb{P}_{\eta\sim q}(f(x+\eta)=A) \ge 1-\alpha$, one has $\mathbb{P}_{\eta\sim q}(f(x+\delta+\eta)=A) > 1/2$ for all $\|\delta\| \le \varepsilon$.

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The condition $\mathbb{P}(f(x+\delta+\eta)=A) > 1/2$ forces $g(x+\delta)=A$, so R_q is the largest radius that can be certified from the confidence level alone, for every base classifier. By the Cohen–Rosenfeld–Kolter bound, $R_{N(0,\sigma^2 I)}(\alpha, \|\cdot\|_2) = \sigma\Phi^{-1}(1-\alpha)$.

Theorem 1.2. *Let q be a Borel probability measure on $\mathbb{R}^d$ with $\mathbb{E}_{\eta\sim q}\|\eta\|_2^2 \le d\sigma^2$, where $\sigma > 0$. Let $0 < \alpha < 1/4$ and*

$$d \;\ge\; \frac{1012\,\log(16/\alpha)}{\alpha^2}.$$

Then for every $2 \le p \le \infty$,

$$R_q(\alpha, \|\cdot\|_p) \;\le\; 8\,\sigma\sqrt{\log(16/\alpha)}\; d^{1/p-1/2}.$$

Corollary 1.3. *Under the assumptions of Theorem 1.2, for every $2 \le p \le \infty$,*

$$R_q(\alpha, \|\cdot\|_p) \;\le\; 25\, R_{N(0,\sigma^2 I)}(\alpha, \|\cdot\|_p),$$

and the ratio of the two bounds tends to at most $8/\sqrt{2} < 5.7$ as $\alpha \to 0$.

The constants are not optimized. Only the second moment of q enters; the noise may be any law, with atoms, heavy tails or strong dependence between coordinates. The proof is an explicit construction: a base classifier that has probability at least $1-\alpha$ of the top class at x and probability at most $1/2$ at a point $x+v$ with $\|v\|_\infty = \varepsilon$. The shift v is a multiple of a sign vector, which is why the ℓ_p bounds follow from the ℓ_∞ one.

2. A criterion for failure

Lemma 2.1. *Let $\|\cdot\|$ be a norm on $\mathbb{R}^d$. Suppose there are a Borel set $B_0 \subset \mathbb{R}^d$ and a vector v with $q(B_0) \ge 1/2$ and $q(v+B_0) \le \alpha$. Then $R_q(\alpha, \|\cdot\|) \le \|v\|$.*

Proof. Fix x and two classes $A \ne B$. Let $f(y) = B$ if $y \in x+v+B_0$ and $f(y) = A$ otherwise. Then $\mathbb{P}(f(x+\eta)=B) = q(v+B_0) \le \alpha$, so $\mathbb{P}(f(x+\eta)=A) \ge 1-\alpha$. At the point $x+v$, $\mathbb{P}(f(x+v+\eta)=B) = q(B_0) \ge 1/2$, so $\mathbb{P}(f(x+v+\eta)=A) \le 1/2$. Hence no $\varepsilon \ge \|v\|$ has the property of Theorem 1.1. □

3. Proof of the theorem

Throughout, $\eta \sim q$, $B_r = \{y : \|y\|_2 \le r\}$, and $\varepsilon > 0$ is a number with $\varepsilon \le \sigma$ to be fixed at the end.

Lemma 3.1 (a thin shell)**.** *There is $\rho \in [2\sigma\sqrt{d},\, 3\sigma\sqrt{d} - \varepsilon\sqrt{d}\,]$ with $q(B_\rho) \ge 3/4$ and*

$$q\big(\rho < \|\eta\|_2 \le \rho + \varepsilon\sqrt{d}\big) \;\le\; \frac{2\varepsilon}{\sigma}.$$

Proof. By Markov's inequality $q(\|\eta\|_2 > 2\sigma\sqrt{d}) \le \mathbb{E}\|\eta\|_2^2/(4d\sigma^2) \le 1/4$, so $q(B_\rho) \ge 3/4$ for every $\rho \ge 2\sigma\sqrt{d}$. The interval $[2\sigma\sqrt{d}, 3\sigma\sqrt{d}]$ contains $N = \lfloor \sigma/\varepsilon \rfloor \ge 1$ disjoint half-open intervals $(\rho_k, \rho_k + \varepsilon\sqrt{d}]$, $\rho_k = 2\sigma\sqrt{d} + k\varepsilon\sqrt{d}$, $0 \le k < N$. The corresponding shells are disjoint, so one of them has mass at most $1/N \le 2\varepsilon/\sigma$, using $\lfloor t \rfloor \ge t/2$ for $t \ge 1$. □

Fix ρ as in Theorem 3.1 and let μ be the law of η conditioned on $\eta \in B_\rho$. For a sign vector $s \in \{-1,1\}^d$ write $X_s(y) = \langle y, s\rangle/\sqrt{d}$. Put

$$T = \sigma\sqrt{18\log 24}, \qquad T' = \sigma\sqrt{18\log(16/\alpha)}.$$

Lemma 3.2 (a direction)**.** *There is $s \in \{-1,1\}^d$ with $\mu(X_s > T) \le 1/3$ and $\mu(X_s < -T') \le \alpha/2$.*

Proof. Let s be uniform on $\{-1,1\}^d$, independent of $\eta \sim \mu$. For fixed $y \in B_\rho$, $\langle y, s\rangle$ is a sum of independent bounded terms $s_i y_i$, and Hoeffding's inequality gives, for $u > 0$,

$$\mathbb{P}_s\big(X_s(y) > u\big) \le \exp\Big(-\frac{u^2 d}{2\|y\|_2^2}\Big) \le \exp\Big(-\frac{u^2}{18\sigma^2}\Big),$$

since $\|y\|_2 \le \rho \le 3\sigma\sqrt{d}$; the same bound holds for $\mathbb{P}_s(X_s(y) < -u)$. Integrating over y, $\mathbb{E}_s\,\mu(X_s > T) \le 1/24$ and $\mathbb{E}_s\,\mu(X_s < -T') \le \alpha/16$. By Markov's inequality in s, $\mathbb{P}_s(\mu(X_s > T) > 1/3) \le 1/8$ and $\mathbb{P}_s(\mu(X_s < -T') > \alpha/2) \le 1/8$, so some s satisfies both bounds. □

Proof of Theorem 1.2. Take s from Theorem 3.2 and set

$$B_0 = \{y \in B_\rho :\ X_s(y) \le T\}, \qquad \varepsilon = \frac{T+T'}{\sqrt{d}}, \qquad v = -\varepsilon s.$$

First, $q(B_0) = q(B_\rho)\,\mu(X_s \le T) \ge \frac{3}{4}\cdot\frac{2}{3} = \frac{1}{2}$.

Next let $y = v + z$ with $z \in B_0$. Then $\|y\|_2 \le \rho + \|v\|_2 = \rho + \varepsilon\sqrt{d}$ and $X_s(y) = X_s(z) - \varepsilon\sqrt{d} \le T - (T+T') = -T'$. So y lies either in the shell $\{\rho < \|y\|_2 \le \rho + \varepsilon\sqrt{d}\}$ or in $B_\rho \cap \{X_s \le -T'\}$, and

$$q(v + B_0) \ \le\ \frac{2\varepsilon}{\sigma} + q(B_\rho)\,\mu(X_s \le -T') \ \le\ \frac{2\varepsilon}{\sigma} + \frac{\alpha}{2},$$

where $\mu(X_s \le -T') \le \alpha/2$ holds by the proof of Theorem 3.2, the Hoeffding bound being the same for the closed event.

It remains to check $2\varepsilon/\sigma \le \alpha/2$ and $\varepsilon \le \sigma$. For $\alpha < 1/4$ we have $\log(16/\alpha) > \log 64$, hence

$$\frac{T+T'}{\sigma} = \sqrt{18}\Big(\sqrt{\log 24} + \sqrt{\log(16/\alpha)}\Big) \le \sqrt{18}\Big(\sqrt{\tfrac{\log 24}{\log 64}} + 1\Big)\sqrt{\log(16/\alpha)} \le 7.952\sqrt{\log(16/\alpha)}.$$

The condition $\varepsilon \le \alpha\sigma/4$ reads $\sqrt{d} \ge 4(T+T')/(\alpha\sigma)$, which holds when $d \ge 16 \cdot 7.952^2\,\alpha^{-2}\log(16/\alpha)$, and $16 \cdot 7.952^2 < 1012$. Then also $\varepsilon \le \sigma$. So $q(v + B_0) \le \alpha$, and Theorem 2.1 gives, since v is ε times a sign vector,

$$R_q(\alpha, \|\cdot\|_p) \le \|v\|_p = \varepsilon\, d^{1/p} = \frac{T+T'}{\sigma}\,\sigma\, d^{1/p-1/2} \le 8\,\sigma\sqrt{\log(16/\alpha)}\, d^{1/p-1/2}.$$ □

Proof of Theorem 1.3. For $p \ge 2$, $\|\delta\|_2 \le d^{1/2-1/p}\|\delta\|_p$, so the Gaussian certifies the ℓ_p radius $\sigma\Phi^{-1}(1-\alpha)\,d^{1/p-1/2}$. The function $\alpha \mapsto 8\sqrt{\log(16/\alpha)}/\Phi^{-1}(1-\alpha)$ is at most 24.19 on $(0, 1/4)$, with the largest values near $\alpha = 1/4$, and tends to $8/\sqrt{2}$ as $\alpha \to 0$ because $\Phi^{-1}(1-\alpha) \sim \sqrt{2\log(1/\alpha)}$. □

4. Remarks

Remark 4.1. The Gaussian value $\sigma\Phi^{-1}(1-\alpha)/\sqrt{d}$ in ℓ_∞ is exact for $N(0, \sigma^2 I)$: the shift $v = \varepsilon(1, \dots, 1)$ and the half-space classifier $f(y) = B$ on $\{\langle y - x, (1,\dots,1)\rangle \ge \sigma\sqrt{d}\,\Phi^{-1}(1-\alpha)\}$ attain it. Theorem 1.2 says that no other law with the same average per-coordinate second moment improves this by more than a constant factor, at any confidence level.

Remark 4.2. The noise level is measured by the average second moment $d^{-1}\mathbb{E}\|\eta\|_2^2$, as in [3]; the bound of [1] is stated for most individual coordinates.

Remark 4.3. The condition $d \ge C\alpha^{-2}\log(1/\alpha)$ comes from the shell step, which needs a shell of relative mass $O(\alpha)$, and is not optimized. Three questions remain: whether the Gaussian is exactly optimal among laws with a given second moment; whether the condition on d can be removed; and what happens for input-dependent smoothing, where the noise law varies with x [6]. The argument above uses a single law q at x and at $x + v$ and does not apply directly to the last case.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Code and data availability

The code and data for this paper are at https://github.com/yspennstate/smoothing-high-confidence. The paper is part of the Math Brain project, whose public repository is https://github.com/yspennstate/math-brain-public [9].

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. In November 2026 Claude models (Anthropic), working in Claude Code, proved the theorem, checked the construction numerically, and wrote the paper. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

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